Variation of Parameters

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: The Wronskian**

For $y_1 = e^x$, $y_2 = e^{2x}$ (solutions of $y'' - 3y' + 2y = 0$), compute the Wronskian $W = y_1y_2' - y_1'y_2$.

$y_1' = e^x$, $y_2' = 2e^{2x}$. $W = e^x(2e^{2x}) - e^x(e^{2x}) = 2e^{3x} - e^{3x} = e^{3x}$.

2 **[Foundational] • Target: The Wronskian**

For $y_1 = e^{3x}$, $y_2 = xe^{3x}$ (solutions of a repeated-root equation), compute W.

$y_1' = 3e^{3x}$, $y_2' = e^{3x} + 3xe^{3x}$. $W = e^{3x}(e^{3x} + 3xe^{3x}) - 3e^{3x}(xe^{3x}) = e^{6x} + 3xe^{6x} - 3xe^{6x} = e^{6x}$.

3 **[Foundational] • Target: The variation of parameters formula**

State the variation of parameters formula for a particular solution of $y'' + p(x)y' + q(x)y = g(x)$, given independent solutions y_1, y_2 of the homogeneous equation with Wronskian W.

$y_p = -y_1\int \frac{y_2 g}{W}\,dx + y_2\int \frac{y_1 g}{W}\,dx$.

4 **[Foundational] • Target: The variation of parameters formula**

Explain why the ODE must first be written in standard form $y'' + p(x)y' + q(x)y = g(x)$ (leading coefficient 1) before applying the variation of parameters formula.

The formula's Wronskian and integrals are derived under the assumption that $g(x)$ is exactly the right side once the leading coefficient of y'' is 1. If the equation is given as $a(x)y'' + b(x)y' + c(x)y = f(x)$ with $a(x) \neq 1$, using $f(x)$ directly in place of $g(x)$ would introduce an extra factor of $a(x)$ and give a wrong particular solution — the equation must first be divided by $a(x)$, giving $g(x) = f(x)/a(x)$.

5 **[Foundational] • Target: A forcing term with no undetermined-coefficients trial form**

Solve $y'' + y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.

Here $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$, $g = \tan x$. Since $\int \cos x \tan x\,dx = \int \sin x\,dx = -\cos x$ and $\int \sin x \tan x\,dx = \int \sec x\,dx - \int \cos x\,dx = \ln|\sec x + \tan x| - \sin x$, we get $y_p = -\cos x[\ln|\sec x + \tan x| - \sin x] + \sin x(-\cos x) = -\cos x \ln|\sec x + \tan x|$.
General solution: $y = C_1\cos x + C_2\sin x - \cos x \ln|\sec x + \tan x|$.

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6 [Foundational] • Target: Comparing with undetermined coefficients

Solve $y'' - y = e^{2x}$ by variation of parameters, then check the particular solution using undetermined coefficients.

$y_1 = e^x$, $y_2 = e^{-x}$, $W = -2$. With $g = e^{2x}$: $\frac{y_2 g}{W} = -\frac{1}{2}e^x$, integral $-\frac{1}{2}e^x$; $\frac{y_1 g}{W} = -\frac{1}{2}e^{3x}$, integral $-\frac{1}{6}e^{3x}$. So $y_p = -e^x(-\frac{1}{2}e^x) + e^{-x}(-\frac{1}{6}e^{3x}) = \frac{1}{2}e^{2x} - \frac{1}{6}e^{2x} = \frac{1}{3}e^{2x}$. Check by undetermined coefficients: try Ae^{2x} (no resonance, since 2 is not a root of $r^2 - 1 = 0$): $4A - A = 1 \Rightarrow A = \frac{1}{3}$. Matches. General solution: $y = C_1 e^x + C_2 e^{-x} + \frac{1}{3}e^{2x}$.

7 [Foundational] • Target: Comparing with undetermined coefficients

Solve $y'' - y = 4$ (a constant forcing) by variation of parameters, and check against the obvious undetermined-coefficients guess $y_p = A$.

$y_1 = e^x$, $y_2 = e^{-x}$, $W = -2$, $g = 4$. $\frac{y_2 g}{W} = -2e^{-x}$, integral $2e^{-x}$; $\frac{y_1 g}{W} = -2e^x$, integral $-2e^x$. So $y_p = -e^x(2e^{-x}) + e^{-x}(-2e^x) = -2 - 2 = -4$. Undetermined coefficients: try $y_p = A$, giving $-A = 4 \Rightarrow A = -4$. Matches. General solution: $y = C_1 e^x + C_2 e^{-x} - 4$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: When to prefer each method

For $y'' + 4y = x^3 - 2x$, which method is more efficient: undetermined coefficients or variation of parameters? Explain.

Undetermined coefficients is far more efficient here: the forcing is a plain polynomial (no resonance issues), so a simple trial $y_p = Ax^3 + Bx^2 + Cx + D$ solves it by matching coefficients — no integration required. Variation of parameters would work too, but requires computing W and two integrals unnecessarily.

9 [Intermediate] • Target: When to prefer each method

For $y'' + y = \csc x$ (for $0 < x < \pi$), which method must be used, and why?

Variation of parameters is required. $\csc x = 1/\sin x$ is not a polynomial, exponential, sine/cosine, or a finite product of these, so undetermined coefficients has no valid trial form; only variation of parameters, which places no restriction on the forcing term, can produce a particular solution.

10 [Intermediate] • Target: Solving $y''+y=\csc x$ in full

Complete the solution of $y'' + y = \csc x$ ($0 < x < \pi$) started above, using $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$.

$\frac{y_2 g}{W} = \frac{\sin x}{\sin x} = 1$, so $\int = x$. $\frac{y_1 g}{W} = \frac{\cos x}{\sin x} = \cot x$, so $\int = \ln|\sin x|$. Then $y_p = -\cos x(x) + \sin x \ln|\sin x| = -x\cos x + \sin x \ln|\sin x|$. General solution: $y = C_1 \cos x + C_2 \sin x - x\cos x + \sin x \ln|\sin x|$.

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11 **[Intermediate]** • **Target: Solving $y''+y=\csc x$ in full**

Verify (by checking the coefficient of the leading term only) that $y_p = -x\cos x$ contributes a term of the correct order when substituted into $y'' + y$.

For $y_p = -x\cos x$: $y_p' = -\cos x + x\sin x$, $y_p'' = \sin x + \sin x + x\cos x = 2\sin x + x\cos x$. So $y_p'' + y_p = 2\sin x + x\cos x - x\cos x = 2\sin x$ — this piece alone contributes $2\sin x$, which combines with the contribution from $\sin x \ln|\sin x|$ to produce the required $\csc x$ on the right side (a full check of the second piece is a longer computation, but this confirms the $-x\cos x$ term is doing real work, not just decoration).

12 **[Intermediate]** • **Target: Variation of parameters with a shifted independent variable**

For the Cauchy–Euler-type substitution used in some ODE courses, $y_1 = x$, $y_2 = x^{-1}$ solve $x^2y'' + xy' - y = 0$ for $x > 0$. Rewrite this equation in the standard form $y'' + p(x)y' + q(x)y = g(x)$ needed for variation of parameters when the right side is $x^2 \cdot 3x = 3x^3$ (i.e., solve $x^2y'' + xy' - y = 3x^3$).

Dividing by x^2: $y'' + \frac{1}{x}y' - \frac{1}{x^2}y = 3x$, so $g(x) = 3x$ (NOT $3x^3$ — the division by the leading coefficient x^2 is exactly the step the earlier question about standard form warns about).

13 **[Intermediate]** • **Target: Variation of parameters with a shifted independent variable**

Continue the problem above: with $y_1 = x$, $y_2 = x^{-1}$, compute $W = y_1y_2' - y_1'y_2$.

$y_1' = 1$, $y_2' = -x^{-2}$. $W = x(-x^{-2}) - 1(x^{-1}) = -x^{-1} - x^{-1} = -2x^{-1}$.

14 **[Intermediate]** • **Target: Variation of parameters with a shifted independent variable**

Finish solving $x^2y'' + xy' - y = 3x^3$ ($x > 0$) using $y_1 = x$, $y_2 = x^{-1}$, $W = -2x^{-1}$, and $g(x) = 3x$ (from the two questions above).

$\frac{y_2 g}{W} = \frac{x^{-1}(3x)}{-2x^{-1}} = -\frac{3x}{2}$, so $\int = -\frac{3x^2}{4}$. $\frac{y_1 g}{W} = \frac{x(3x)}{-2x^{-1}} = -\frac{3x^3}{2}$, so $\int = -\frac{3x^4}{8}$. Then $y_p = -x\left(-\frac{3x^2}{4}\right) + x^{-1}\left(-\frac{3x^4}{8}\right) = \frac{3x^3}{4} - \frac{3x^3}{8} = \frac{3x^3}{8}$. General solution: $y = C_1x + C_2x^{-1} + \frac{3x^3}{8}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Variation of parameters with a shifted independent variable** • **Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

Check the particular solution $y_p = \frac{3}{8}x^3$ found above by substituting directly into $x^2y'' + xy' - y = 3x^3$.

$y_p' = \frac{9}{8}x^2$, $y_p'' = \frac{9}{4}x$. Then $x^2y_p'' + xy_p' - y_p = x^2\left(\frac{9}{4}x\right) + x\left(\frac{9}{8}x^2\right) - \frac{3}{8}x^3 = \frac{9}{4}x^3 + \frac{9}{8}x^3 - \frac{3}{8}x^3 = \frac{18}{8}x^3 + \frac{9}{8}x^3 - \frac{3}{8}x^3 = \frac{24}{8}x^3 = 3x^3$. Matches.

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16 [Challenging] • Target: A summary comparison • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

List two advantages of variation of parameters over undetermined coefficients.

(1) It works for ANY continuous forcing term $g(x)$, not just polynomials/exponentials/sines-cosines and their products — it handles $\tan x$, $\sec x$, $1/x$, $\ln x$, and more. (2) It gives a single unified formula/procedure, with no need to guess a trial form or check for resonance case-by-case.

17 [Challenging] • Target: A summary comparison • Cross-Lesson: 1.01 Separable Equations

List one advantage of undetermined coefficients over variation of parameters, when it applies.

It is purely algebraic — matching coefficients — with no integration required, so it is typically much faster when the forcing term is one of its supported families (and the resulting integrals in variation of parameters can themselves be difficult or impossible to evaluate in closed form).

18 [Challenging] • Target: A summary comparison • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

True or false, with a one-sentence justification: "Variation of parameters requires knowing the general solution of the associated homogeneous equation."

True — the formula uses two independent homogeneous solutions y_1, y_2 and their Wronskian W directly, so the homogeneous equation must be solved first, exactly as with undetermined coefficients (which also needs y_c to check for resonance).

19 [Challenging] • Target: A summary comparison • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

For $y'' + 4y = 8\tan 2x$ $(-\frac{\pi}{4} < x < \frac{\pi}{4})$, which method must be used, and what are y_1, y_2, and W?

Variation of parameters is required, since $\tan 2x$ is outside the undetermined-coefficients family. From $r^2 + 4 = 0$: $y_1 = \cos 2x$, $y_2 = \sin 2x$. $W = y_1y_2' - y_1'y_2 = \cos 2x(2\cos 2x) - (-2\sin 2x)\sin 2x = 2\cos^2 2x + 2\sin^2 2x = 2$.

20 [Challenging] • Target: A summary comparison • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

In one sentence, summarize when a student should reach for variation of parameters instead of undetermined coefficients.

Reach for variation of parameters whenever the forcing term $g(x)$ is not a polynomial, exponential, sine/cosine, or product of these (e.g., it involves $\tan$, $\sec$, $\csc$, $\ln$, or a quotient like $1/x$) — otherwise undetermined coefficients is faster.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Wronskian

Find a particular solution of $y'' - 6y' + 8y = 2e^{4x}$ (note: 4 is a root) using variation of parameters.

a. $xe^{4x} - \frac{e^{4x}}{2}$ ✓

b. $-xe^{4x} + \frac{e^{4x}}{2}$

c. $2xe^{4x} - e^{4x}$

d. $\frac{xe^{4x}}{2} - \frac{e^{4x}}{4}$

Homogeneous: $r^2 - 6r + 8 = 0$ has roots $r = 4, 2$, so $y_1 = e^{4x}$, $y_2 = e^{2x}$, $W = -2e^{6x}$. Unlike undetermined coefficients, no resonance check or modification is needed: $u_1 = x$, $u_2 = -\frac{e^{2x}}{2}$, so $y_p = xe^{4x} - \frac{e^{4x}}{2}$.

22 [Foundational] • Target: The variation of parameters formula

For $y'' + 4y = g(x)$, with $y_1 = \cos 2x$, $y_2 = \sin 2x$ ($W = 2$), which is the correct formula for $u_2'(x)$ in variation of parameters?

a. $\frac{g(x)\sin 2x}{2}$

b. $\frac{g(x)\cos 2x}{2}$ ✓

c. $-\frac{g(x)\sin 2x}{2}$

d. $g(x)\cos 2x$

Theorem 5.1 gives $u_2' = \frac{y_1 g}{aW}$; with $y_1 = \cos 2x$, $a = 1$, $W = 2$, this is $u_2' = \frac{g(x)\cos 2x}{2}$.

23 [Foundational] • Target: A forcing term with no undetermined-coefficients trial form

Consider the differential equation $y'' - xy = 0$. We are looking for a power series solution of the form $y(x) = \sum_{n=0}^{\infty} a_n x^n$ around the ordinary point $x = 0$. Find the coefficient a_2 in terms of a_0 and a_1.

a. $a_2 = a_0$

b. $a_2 = a_1$

c. $a_2 = 0$ ✓

d. $a_2 = a_0 + a_1$

We assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} a_n x^n$. Then $y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ and $y''(x) = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$. Substituting these into the differential equation $y'' - xy = 0$, we get: $\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - x\sum_{n=0}^{\infty} a_n x^n = 0$ $\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=0}^{\infty} a_n x^{n+1} = 0$. To compare coefficients, we need the powers of x to be the same. Let $k = n - 2$ in the first sum, so.

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24 [Foundational] • Target: A repeated root, with forcing dividing by x

Find a particular solution of $y'' + 16y = -\sec 4x$ (on an interval where $\sec 4x$ is continuous) using variation of parameters.

a. $\frac{x\sin(4x)}{4} + \frac{\log(\cos(4x))\cos(4x)}{16}$

b. $-\frac{x\sin(4x)}{4} - \frac{\log(\cos(4x))\cos(4x)}{16}$ ✓

c. $-\frac{x\sin(4x)}{2} - \frac{\log(\cos(4x))\cos(4x)}{8}$

d. $-\frac{x\sin(4x)}{8} - \frac{\log(\cos(4x))\cos(4x)}{32}$

Homogeneous: $r^2 + 16 = 0$ has roots $r = 0 \pm 4i$, so $y_1 = \cos 4x$, $y_2 = \sin 4x$, $W = 4$. Solving for u_1, u_2 and substituting back: $y_p = -\frac{x\sin(4x)}{4} - \frac{\log(\cos(4x))\cos(4x)}{16}$.

25 [Foundational] • Target: Comparing with undetermined coefficients

For the classical RK4 method, dividing the step size h by 10 reduces the global error by roughly what factor?

a. 10

b. 10000 ✓

c. 100

d. 20

Global error $\propto h^4$, so the error shrinks by about $10^4 = 10000$.

26 [Foundational] • Target: When to prefer each method

For Euler's method, dividing the step size h by 10 reduces the global error by roughly what factor?

a. 10 ✓

b. 100

c. 10000

d. 20

Global error $\propto h^1$, so the error shrinks by about $10^1 = 10$.

27 [Foundational] • Target: Solving $y''+y=\csc x$ in full

For the improved Euler (Heun) method, dividing the step size h by 3 reduces the global error by roughly what factor?

a. 3

b. 9 ✓

c. 27

d. 81

Global error $\propto h^2$, so the error shrinks by about $3^2 = 9$.

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28 [Foundational] • Target: Variation of parameters with a shifted independent variable

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate the value of $y(0.2)$.

a. 1.20

b. **1.21** ✓

c. 1.22

d. 1.23

The Euler method formula is given by $y_{n+1} = y_n + h \cdot f(x_n, y_n)$, where $f(x, y) = \frac{dy}{dx}$. We are given $f(x, y) = x + y$, $y(0) = 1$, and $h = 0.1$. We want to approximate $y(0.2)$. Step 1: Calculate y_1 at $x_1 = 0 + h = 0.1$. $x_0 = 0$, $y_0 = 1$. $y_1 = y_0 + h \cdot f(x_0, y_0) = 1 + 0.1 \cdot (0 + 1) = 1 + 0.1 \cdot 1 = 1.1$. So, $y(0.1) \approx 1.1$. Step 2: Calculate y_2 at $x_2 = 0.1 + h = 0.2$. $x_1 = 0.1$, $y_1 = 1.1$..

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A summary comparison

Consider the system of linear differential equations given by: $\frac{dx}{dt}, = x + 2y; \frac{dy}{dt}, = 3x + 2y$ If the solution to this system is of the form $x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$ and $y(t) = k_1 c_1 e^{\lambda_1 t} + k_2 c_2 e^{\lambda_2 t}$, what is the value of the larger eigenvalue λ?

a. **6** ✓

b. 5

c. 4

d. 3

The given system can be written in matrix form as $\mathbf{x}'(t) = A\mathbf{x}(t)$, where $A = (1, 2; 3, 2)$ and $\mathbf{x}(t) = (x(t); y(t))$. To find the eigenvalues λ, we solve the characteristic equation $\det(A - \lambda I) = 0$, where I is the identity matrix. $\det(1 - \lambda, 2; 3, 2 - \lambda) = 0$ $(1 - \lambda)(2 - \lambda) - (2)(3) = 0$ $2 - \lambda - 2\lambda + \lambda^2 - 6 = 0$ $\lambda^2 - 3\lambda - 4 = 0$ Factoring the quadratic equation.

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30 [Intermediate] • Target: The Wronskian

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.

a. 1.200

b. **1.210** ✓

c. 1.230

d. 1.220

The Euler method uses the formula $y_{n+1} = y_n + h \cdot f(x_n, y_n)$, where $f(x, y) = x + y$ and $h = 0.1$. Starting with $x_0 = 0$ and $y_0 = 1$: Step 1: Approximate $y(0.1)$. $y_1 = y_0 + h \cdot f(x_0, y_0)$ $y_1 = 1 + 0.1 \cdot (0 + 1)$ $y_1 = 1 + 0.1 \cdot 1$ $y_1 = 1.1$ So, $x_1 = 0.1$ and $y_1 = 1.1$. Step 2: Approximate $y(0.2)$. $y_2 = y_1 + h \cdot f(x_1, y_1)$ $y_2 = 1.1 + 0.1 \cdot (0.1 + 1.1)$ $y_2 = 1.1 + 0.1 \cdot (1.2)$ $y_2 = 1.1 + 0.12$ $y_2 = 1.22$ Therefore, the approximation for $y(0.2)$ is.

31 [Intermediate] • Target: The variation of parameters formula

Consider the initial value problem $y'' + 4y = \delta(t - \pi)$ with $y(0) = 0$ and $y'(0) = 0$. Which of the following represents the Laplace Transform of the solution, $Y(s) = \mathcal{L}\{y(t)\}$?

a. $\frac{e^{-\pi s}}{s^2 + 4}$ ✓

b. $\frac{1}{s^2 + 4}$

c. $\frac{e^{-\pi s}}{s^2}$

d. $\frac{e^{-\pi s}}{s^2 - 4}$

Taking the Laplace transform of the differential equation, we have $s^2Y(s) - sy(0) - y'(0) + 4Y(s) = \mathcal{L}\{\delta(t - \pi)\}$. Given the initial conditions $y(0) = 0$ and $y'(0) = 0$, this simplifies to $s^2Y(s) + 4Y(s) = e^{-\pi s}$. Factoring out $Y(s)$, we get $Y(s)(s^2 + 4) = e^{-\pi s}$. Therefore, the Laplace transform of the solution is $Y(s) = \frac{e^{-\pi s}}{s^2 + 4}$.

32 [Intermediate] • Target: A forcing term with no undetermined-coefficients trial form

Solve $\frac{dy}{dx} = \frac{y \ln x}{x(y^2 + 1)}$ for $x > 0, y > 0$, with $y(1) = 1$. What is the numerical value of $y^2 + 2\ln y$ when $x = e^2$?

a. 5 ✓

b. 6

c. 4

d. 3

Separable: $\frac{y^2 + 1}{y}\,dy = \frac{\ln x}{x}\,dx \Rightarrow \left(y + \frac{1}{y}\right)dy = \ln x\, d(\ln x)$. Integrating gives $\frac{1}{2}y^2 + \ln y = \frac{1}{2}(\ln x)^2 + C$. Using $x = 1, y = 1$ yields $\frac{1}{2} = 0 + C \Rightarrow C = 1/2$. Multiplying by 2 gives $y^2 + 2\ln y = (\ln x)^2 + 1$. At $x = e^2$, $\ln x = 2$, so $y^2 + 2\ln y = 2^2 + 1 = 5$.

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33 [Intermediate] • Target: A repeated root, with forcing dividing by x

Solve $y' + 2y = 4t$ with $y(0) = y_0$. For what value of y_0 does the solution contain no exponential transient term?

a. 0

b. -1 ✓

c. 1

d. $-\frac{1}{2}$

$\mu = e^{2t}$. Multiplying gives $(ye^{2t})' = 4te^{2t}$. Integrating by parts yields $ye^{2t} = 2te^{2t} - e^{2t} + C \Rightarrow y(t) = 2t - 1 + Ce^{-2t}$. For the exponential term to vanish, we must have $C = 0$. Setting $y(0) = y_0$ gives $y_0 = 2(0) - 1 + C = -1 + 0 = -1$.

34 [Intermediate] • Target: Comparing with undetermined coefficients

Solve $\frac{dy}{dx} = \frac{x^2+2}{y^2-1}$ with $y(0) = 2$ for $y > 1$. What is the value of $\frac{1}{3}y^3 - y$ when $x = 3$?

a. 15

b. $\frac{49}{3}$

c. $\frac{47}{3}$ ✓

d. $\frac{45}{3}$

Separable: $(y^2 - 1)\,dy = (x^2 + 2)\,dx$. Integrating gives $\frac{1}{3}y^3 - y = \frac{1}{3}x^3 + 2x + C$. Using $y(0) = 2$ yields $\frac{8}{3} - 2 = \frac{2}{3} = C$. At $x = 3$, $\frac{1}{3}y^3 - y = \frac{1}{3}(27) + 2(3) + \frac{2}{3} = 9 + 6 + \frac{2}{3} = \frac{47}{3}$.

35 [Intermediate] • Target: When to prefer each method

Given $\frac{dy}{dx} = \frac{x^2+1}{y^2+1}$ with $y(0) = 0$, what is the numerical value of $\frac{1}{3}y^3 + y$ when $x = 3$?

a. 12 ✓

b. 15

c. 10

d. 9

Separable: $(y^2 + 1)\,dy = (x^2 + 1)\,dx$. Integrating yields $\frac{1}{3}y^3 + y = \frac{1}{3}x^3 + x + C$. With $y(0) = 0$, $0 = C$. Thus $\frac{1}{3}y^3 + y = \frac{1}{3}x^3 + x$. At $x = 3$, $\frac{1}{3}(27) + 3 = 9 + 3 = 12$.

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36 [Intermediate] • Target: Solving $y''+y=\csc x$ in full

Which recurrence do the coefficients of $y = \sum a_n x^n$ satisfy for $y'' - xy' - y = 0$?

a. $a_{n+2} = \frac{a_n}{(n+2)(n+1)}$

b. $a_{n+2} = \frac{a_n}{n+2}$ ✓

c. $a_{n+2} = -\frac{a_n}{n+2}$

d. $a_{n+2} = \frac{(n+1)a_n}{n+2} \cdot 2$

Substitute $y'' = \sum (n+2)(n+1)a_{n+2}x^n$ and shift indices so every sum runs over x^n; setting each coefficient to zero gives $a_{n+2} = \frac{a_n}{n+2}$.

37 [Intermediate] • Target: Variation of parameters with a shifted independent variable

Use the improved Euler (Heun) method with $h = \frac{1}{2}$ to approximate $y(\frac{1}{2})$ for $y' = y - x$, $y(0) = 2$.

a. $\frac{25}{8}$ ✓

b. 3

c. 2

d. $\frac{9}{2}$

$y_{n+1} = y_n + \frac{h}{2}[f(x_n, y_n) + f(x_n + h,\ y_n + hf(x_n, y_n))]$, applied 1 time(s) from $(0, 2)$, gives $\frac{25}{8}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A summary comparison • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

On $[0, 1]$, Euler's method with $h = 0.1$ gives an error of about $\frac{1}{50}$. Roughly how many steps are needed for an error of $\frac{1}{500}$?

a. 100 ✓

b. 1000

c. 200

d. 50

The error scales like h^1; reducing it by 10 needs h smaller by 10, i.e. $h = \frac{1}{100}$, which is 100 steps.

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39 [Challenging] • Target: The Wronskian • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

On $[0, 1]$, the improved Euler method with $h = 0.1$ gives an error of about $\frac{1}{50}$. Roughly how many steps are needed for an error of $\frac{1}{200}$?

a. 40

b. 11

c. 30

d. 20 ✓

The error scales like h^2; reducing it by 4 needs h smaller by 2, i.e. $h = \frac{1}{20}$, which is 20 steps.

40 [Challenging] • Target: The variation of parameters formula • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Use the classical RK4 method with $h = \frac{1}{2}$ to approximate $y(\frac{3}{2})$ for $y' = x - y$, $y(1) = 2$.

a. $\frac{3}{2}$

b. $\frac{329}{192}$ ✓

c. 2

d. $\frac{7}{4}$

$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$, applied 1 time(s) from $(1, 2)$, gives $\frac{329}{192}$.

41 [Challenging] • Target: A forcing term with no undetermined-coefficients trial form • Cross-Lesson: 1.01 Separable Equations

For $y' = -y$, $y(0) = 1$, Euler's method with $h = \frac{1}{4}$ gives $y(1) \approx \frac{81}{256}$. What is the actual error $y(1) - y_4$?

a. $\frac{81}{256} - e^{-1}$

b. $-\frac{81}{256} + e^{-1}$ ✓

c. $\frac{81}{256}$

d. e^{-1}

The exact solution is $y = e^{-x}$, so $y(1) = e^{-1}$ and the error is $e^{-1} - \frac{81}{256}$.

42 [Challenging] • Target: A repeated root, with forcing dividing by x • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

What is the global truncation error of the improved Euler (Heun) method?

a. $O(h^2)$ ✓

b. $O(h)$

c. $O(h^3)$

d. $O(h^4)$

The improved Euler (Heun) method is a method of order 2: its global error is $O(h^2)$.

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43 [Challenging] • Target: Comparing with undetermined coefficients • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

What is the global truncation error of the classical RK4 method?

a. $O(h)$

b. $O(h^4)$ ✓

c. $O(h^2)$

d. $O(h^3)$

The classical RK4 method is a method of order 4: its global error is $O(h^4)$.

44 [Challenging] • Target: When to prefer each method • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Use Euler's method with $h = \frac{1}{4}$ to approximate $y(\frac{1}{4})$ for $y' = -y$, $y(0) = 3$.

a. $\frac{75}{32}$

b. 3

c. $\frac{3}{2}$

d. $\frac{9}{4}$ ✓

$y_{n+1} = y_n + hf(x_n, y_n)$, applied 1 time(s) from $(0, 3)$, gives $\frac{9}{4}$.

45 [Challenging] • Target: Solving $y''+y=\csc x$ in full • Cross-Lesson: 1.01 Separable Equations

What is the global truncation error of Euler's method?

a. $O(h^2)$

b. $O(h^3)$

c. $O(h^4)$

d. $O(h)$ ✓

Euler's method is a method of order 1: its global error is $O(h)$.

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